

1. GENERAL INFORMATION & DELIVERABLE LINKS

PocketReceipt

Course: Cross-Platform Mobile App Development (VKU) - Mini-Project 3

Mini-Project Title	OCR Expense Tracker & Receipt Parser
Team / Student	Tran Lam
Student ID	Not provided
Role / Contribution	Individual project - architecture, implementation, testing - 100%
Submission Date	08/10/2026

Deliverable links

LIVE DEMO	https://pocketreceipt-demo.onrender.com
GITHUB	https://github.com/it2kvku/pocketreceipt-ocr-expense-tracker
APK + VIDEO	https://github.com/it2kvku/pocketreceipt-ocr-expense-tracker/releases/tag/v1.0.0

2. Feature implementation checklist

1	Camera capture & crop Live viewfinder, flash, tap focus, framing overlay, crop.	COMPLETE
2	On-device OCR Bundled ML Kit Latin recognizer; no cloud OCR.	COMPLETE
3	Regex heuristics Merchant, validated date and ranked VND total.	COMPLETE
4	Review & lifecycle Editable fields; add, search, filter, edit, delete.	COMPLETE
5	Local persistence SQLite plus private receipt and thumbnail files.	COMPLETE
6	Canvas visualization Animated interactive donut and weekly bars.	COMPLETE

3. TECHNICAL ARCHITECTURE & PROJECT STRUCTURE

From receipt pixels to a reviewed expense.

A modular local-first pipeline with explicit review and no application server.

CAPTURE / CROP

Camera or gallery - orientation - crop - private JPEG

ON-DEVICE OCR

ML Kit Latin recognizer - spatial rows - raw text

PARSE / REVIEW

Regex candidates - ranked totals - editable confirmation

PERSIST / VISUALIZE

SQLite + thumbnails - calendar summaries - CustomPainter

Project structure

domain/ contains the Expense model, categories and deterministic parser. services/ owns ML Kit recognition and disposal. data/ selects SQLite/private files on Android or preferences on web. ui/ contains capture, crop, review, overview, history and custom chart painters.

State, data and exception handling

The app reloads stored expenses into an in-memory list, derives calendar summaries, and refreshes after each write. Schema v1 stores ID, merchant, positive integer VND amount, local ISO date, category, receipt path and OCR text. Camera, OCR and database errors appear as user messages. Cancelled scans and deleted records remove cached files; Android backup is disabled.

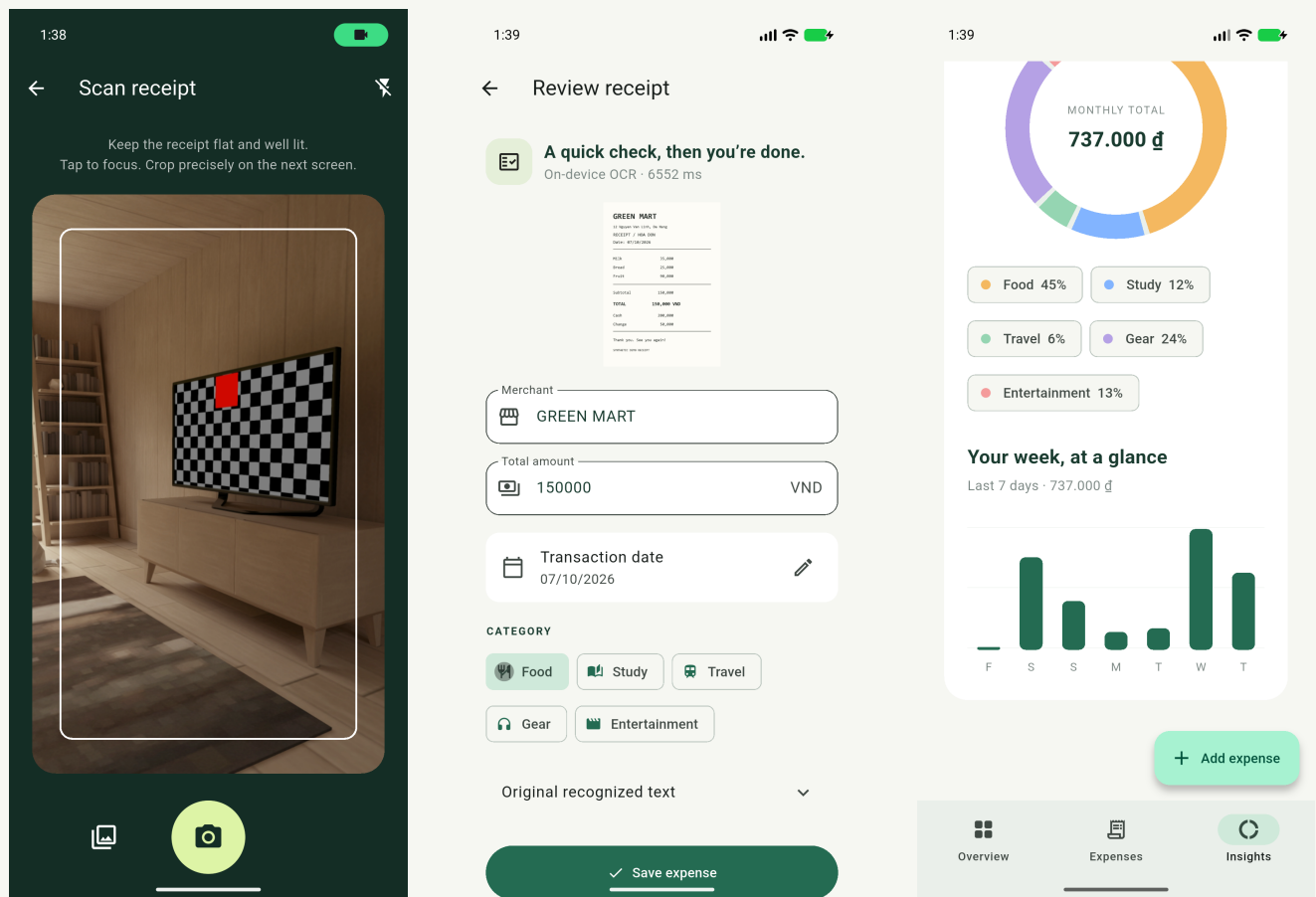
Parser acceptance strategy

Grand-total labels outrank generic totals. Subtotal, VAT, cash, discount and change lines are excluded. Dates are calendar-validated; missing or competing candidates raise review warnings. Every extracted field remains editable.

4. EMPIRICAL EVIDENCE & SCREENSHOTS

The Android workflow running end to end.

Actual emulator captures. The receipt fixture is synthetic; OCR uses the native ML Kit model.



01 Capture / import

Camera controls and framing overlay.

02 OCR review

Merchant, amount and date remain editable.

03 Spending insights

Donut and weekly bars drawn on canvas.

MEASURED RESULT

The final release APK was installed on the Android emulator. Importing the synthetic receipt into native ML Kit returned GREEN MART, 150000 VND and 07/10/2026 correctly, with a displayed OCR-call duration of 6552 ms. Saving and force-stopping/relaunching preserved the expense in SQLite. The 150-second video records this flow and then adds labelled sample data to exercise interactive charts and history.

5. TECHNICAL CHALLENGES & RESOLUTIONS

Failures turned into release safeguards.

The two main bottlenecks appeared during release integration and real OCR output.

01 ML Kit failed after release shrinking

R8 removed Firebase component registrar constructors required at runtime. Release rules now preserve ComponentRegistrar implementations while suppressing only missing optional CJK recognizer classes. A release APK was rebuilt and exercised with actual OCR.

02 OCR columns lost receipt row meaning

ML Kit can return labels and amounts in separate blocks, so plain concatenation paired the wrong values. Bounding boxes are now sorted vertically, grouped into rows by text height, then ordered left-to-right before parsing.

Verification

29 automated tests passed and flutter analyze reported no issues. Tests cover VND separators, invalid dates and amounts, Vietnamese and split-line totals, spatial row reconstruction, ambiguity warnings, category suggestions, SQLite lifecycle, review validation and empty charts. The release APK was signature-verified, installed, used for native OCR, saved to SQLite, force-stopped and relaunched successfully.

Measured limits

The emulator displayed 6552 ms for the demonstrated OCR call, so the sub-100 ms target is not verified. Physical camera focus, flash, thermal print quality and varied merchants still require device testing. The current parser supports integer VND and Latin-script recognition; extracted fields require review.

Reproduce

flutter pub get - flutter analyze - flutter test - flutter build apk --release. Minimum Android SDK: 24. The coursework APK is development-signed for sideloading; store publication requires a private upload key.